

Boolean Algebra:

OR rules //

$$\left\{ \begin{array}{l} 0+0=\underline{0} \\ 1+0=\underline{1} \\ 0+1=\underline{1} \\ 1+1=\underline{1} \end{array} \right.$$

$$A, 0, 1$$

$$\rightarrow A+0=A \checkmark$$

$$\begin{array}{cc} 0 & 0 \\ 1 & 1 \end{array}$$

$$\rightarrow A+1=1$$

$$\begin{array}{cc} 0 & 1 \\ 1 & 1 \end{array}$$

$$\rightarrow A+A=A$$

$$A=0 \quad 0+0=0 \leftarrow A$$

$$A=1 \quad 1+1=1 \leftarrow A$$

$$\bar{A}=1$$

$$A=0 \rightarrow$$

$$A=1 \rightarrow \bar{A}=0$$

$$\rightarrow A+\bar{A}=1$$

$$A=0 \quad 0+1=1$$

$$A=1 \quad 1+0=1$$

$$\text{Identity law, } A+0=A$$

$$\text{Not } \rightarrow \begin{cases} 1, T \rightarrow 0, F \\ 0, F \rightarrow 1, T \end{cases}$$

And Rules

$$\left\{ \begin{array}{l} 0 \cdot 0 = 0 \\ 1 \cdot 0 = 0 \\ 0 \cdot 1 = 0 \\ 1 \cdot 1 = \underline{1} \end{array} \right. \leftarrow 1 \}$$

$$\rightarrow A \cdot 0 = 0$$

$$\rightarrow A \cdot 1 = A$$

$$\rightarrow A \cdot A = A$$

$$\left. \begin{array}{l} A=0 \quad 0 \cdot 0 = 0 \\ A=1 \quad 1 \cdot 1 = 1 \end{array} \right\} A$$

$$\rightarrow A \cdot \bar{A} = 0$$

$$1 \cdot 0 = 0$$

$$A=0, \bar{A}=1$$

$$0 \cdot 1 = 0$$

$$A \cdot 1 = A$$

$$A \cdot A = A$$

Idempotent law, $\underline{A + A = A}$ $\underline{A \cdot A = A}$
Null law, $\underline{A + 1 = 1}$ $\underline{A \cdot 0 = 0}$
Inverse law, $\underline{A + \bar{A} = 1}$ $\underline{A \cdot \bar{A} = 0}$
Inversion law, $\underline{\bar{\bar{A}} = A}$ $\begin{matrix} 1 + 0 = 1 \\ 0 + 1 = 1 \end{matrix}$

$\bar{A}$ negation ($\bar{A}$)
 of not A
 $(\bar{A}) = \neg A$

$\bar{A} = \neg A$
 $(\bar{A})$

Commutative law,
 $\underline{A + B = B + A}$, $\underline{A \cdot B = B \cdot A}$

Associative law,
 $\underline{A + (B + C) = (A + B) + C}$
 $\underline{A \cdot (B \cdot C) = (A \cdot B) \cdot C}$

Distributive law,
 $\underline{A(B + C) = AB + AC}$ $\underline{(A + B)(A + C) = A + BC}$

De Morgan's Law:

$\underline{\overline{A + B} = \bar{A} \cdot \bar{B}}$
 $A + B = P$

$A + B = 0$

$\underline{\overline{A + B} = 1}$

$\underline{\overline{A \cdot B} = \bar{A} + \bar{B}}$
 $\underline{\overline{A + B} = \bar{A} \cdot \bar{B}}$

Simplification of complex Boolean Expressions

$* (A + B)(A + C) = A + BC$

$$L.H.S = (A+B)(A+C)$$

$$= AA + AC + AB + BC$$

$$= \underline{A} + \underline{AC} + \underline{AB} + \underline{BC}$$

$$= \underline{A(1+C)} + \underline{B(A+C)} + AB + BC$$

$$= \underline{A \cdot 1^1} + AB + BC$$

$$\underline{A} + AB + BC = \underline{A(\bar{1} + \underline{B})} + BC$$

$$= \underline{A \cdot 1} + BC$$

$$= \underline{R.H.S}$$

$$* \quad AB(\underline{\bar{A} + BC})$$

$$= AB(\underline{\bar{A}} \cdot \underline{\bar{B}C})$$

$$= AB(A(\bar{B} + \bar{C}))$$

$$= AB(\underline{A\bar{B}} + \underline{A\bar{C}})$$

$$= \underline{AB A\bar{B}} + \underline{AB A\bar{C}}$$

$$= \underline{\overset{A}{A} \underline{\bar{A}} B \bar{B}} + \underline{\overset{A}{A} \underline{\bar{A}} B \bar{C}} = AB\bar{C}$$

$$1 \cdot 0$$

$$\underline{AB(\bar{A} + \underline{BC})} = \underline{AB\bar{C}} \leftarrow$$

$$\underline{A = 0}$$

$$\underline{B = 1}$$

$$\underline{C = 1}$$

$$L.H.S = 0.1(\underline{1+1})$$

$$= \underline{0.0.0} \leftarrow$$

$$= \underline{0}$$

$$R.H.S = 0.1.0 = \underline{0}$$

$$* (\underbrace{A\bar{B}}_{\sim} (\underbrace{C+BD})_{\sim} + \bar{A}\bar{B})C$$

$$\begin{aligned}\bar{B}.B &= 0 \\ \Rightarrow 0.1 &= 0 \\ \underline{1.0} &= 0\end{aligned}$$

$$= (\underbrace{A\bar{B}C}_{\sim} + \underbrace{A\bar{B}BD}_{\sim} + \bar{A}\bar{B})C$$

$$= (\underbrace{A\bar{B}C}_{\sim} + \underbrace{\bar{A}\bar{B}}_{\sim})C$$

$$\underline{\underline{\tilde{C}.C = C}}$$

$$= A\bar{B}\underline{C} + \bar{A}\bar{B}C$$

$$= \boxed{\underbrace{A\bar{B}C}_{\sim} + \underbrace{\bar{A}\bar{B}C}_{\sim}}$$

$$\boxed{A.0.D}$$

$$= 0$$

$$= \underline{\bar{B}C} (\underbrace{A+\bar{A}}_{\sim})$$

$$\underline{A+\bar{A}} = 1$$

$$= \underline{\bar{B}C} 1$$

$$A.\bar{A} = 0$$

$$\boxed{\overline{AB}}$$

A and B

$$= \boxed{\text{not } A \text{ or not } B}$$

Dem

$$\underline{\overline{AB}} = \boxed{\bar{A} + \bar{B}}$$

$$\swarrow$$

$$\text{not } A \text{ and } B = \text{not } A \text{ or not } B$$

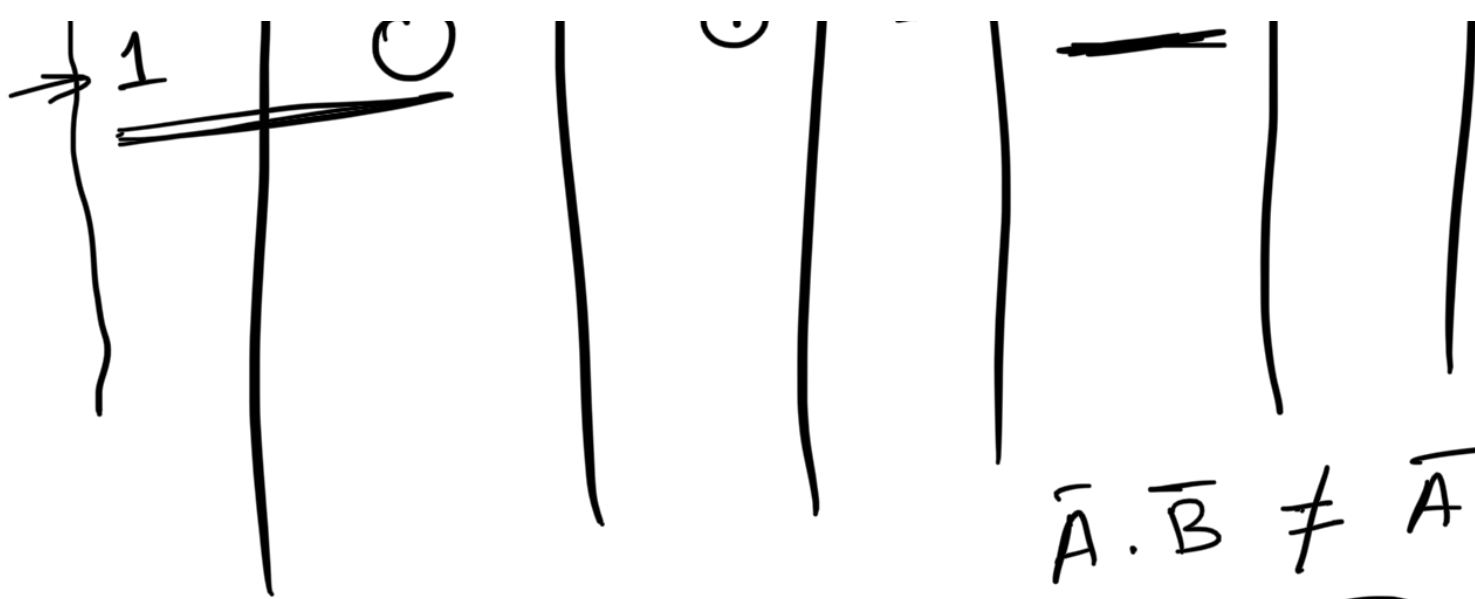
$\bar{A}$ and $\bar{B}$

$\bar{A}$ and $\bar{B}$

AB

$$* \bar{A}.\bar{B} \quad \overline{AB}$$

A	B	$\bar{A}$	$\bar{B}$	$\overline{\bar{A}.\bar{B}}$	AB	$\underline{\underline{\overline{AB}}}$
1	1	0	0	0	1	0
1	0	0	1	0	0	1



$$\bar{A} \cdot \bar{B} \neq \overline{AB}$$

question
↙

Ans
↙

$$\frac{(A+B)C}{1 \quad 1 \quad 0} = \frac{AB+AC}{1 \quad 1 \quad 1 \quad 0}$$

$\Rightarrow$

A, B, C

* $\frac{1+A=1}{A=1}$

$1+1 = \underline{1}$

$A=0$

$1+0 = \underline{1}$

* $\frac{(A+B)(\bar{A}+B)\bar{B}}{AB\bar{B} + B\bar{B}A + B\bar{B}} = (AB + B\bar{A} + B)\bar{B}$

$= AB\bar{B} + B\bar{B}A + B\bar{B}$

$\frac{A+B+B+C}{A+B+C}$

$\boxed{BA=0}$

$\underline{AB(\bar{A}+B)}$

$$A \cdot \bar{A} = 0$$

$$\underline{A \cdot 0 = 0}$$

$$\underline{\bar{A}B(A+B)}$$

$$\bar{A}B(\bar{A}+B)$$

$$\underline{\bar{A}B\bar{A}} + \underline{\bar{A}BB}$$

$$\bar{A}B(\bar{A}+B)$$

$$= \bar{A}\bar{A} + \bar{B}\bar{A} + BB$$

$$\underline{\bar{A}B(\bar{A}+B)}$$

$$\bar{A}B(\bar{A}+B) = \bar{A}\bar{A} +$$

$$\underline{\bar{A}\bar{A} + 1}$$

$$\bar{A}B(\bar{A}+B)$$

$$= (\bar{A}+B)(\bar{A}+B)$$

$$= \bar{A}\bar{A} + \bar{A}B + \bar{A}\bar{B} + \bar{B}B$$

$$= \bar{A} + \bar{A}B + \bar{A}\bar{B} + \underline{\bar{B}B}$$

$$= \bar{A} + \bar{A}(B+\bar{B})$$

$$= \bar{A} + \bar{A}(1)$$

$$= \bar{A} + \bar{A} = \underline{\bar{A}}$$

~~$$\bar{A} + \bar{B} = \overline{AB}$$~~

$$\bar{A}B = \bar{A} * \bar{B} \quad \text{De Morgan's}$$

$$\{ \overline{A+B} = \overline{A} \cdot \overline{B} \}$$

sets

$U \cap -$

De Morgan's law

$$* A \cdot 1 = A$$

$$\underline{A \cdot 1 = A}$$

$$\begin{aligned} &\swarrow \\ &A=0, \quad 0 \cdot 1 = 0 = \underline{A} \\ &\searrow \\ &A=1, \quad 1 \cdot 1 = 1 = \underline{A} \end{aligned}$$

$$\underline{A \cdot A = A}$$

3

$$\begin{aligned} A(A+C) + C &= \cancel{AA} + AC + C \\ &= A + C(\underline{1+A}) \\ &= A + C \cdot 1 = \underline{A+C} \end{aligned}$$

A

$$\left\{ \begin{aligned} 1+A &= 1 \\ A+A &= A \\ A \cdot A &= A \\ A+\bar{A} &= 1 \\ \underline{A \cdot \bar{A}} &= \underline{0} \end{aligned} \right.$$

+ De Morgan's law

$$\rightarrow \bar{A} \cdot A$$

$$A=0,$$

$$1 \cdot 0 = 0$$

$$A=1$$

$$- 0 \cdot 1 = 0$$

$$\underline{\bar{A} \cdot A = 0}$$

+ find if have anything in common

$$\underline{A(1+A)} \\ \underline{(A+\bar{A})}$$

